

Foundations for
programming languages
5: An operational semantics of
a small subset of C

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Today's topic

- Give an operational semantics to a tiny subset of the language C.
 - We use an operational semantics called natural semantics or structural operational semantics.

Arithmetic expressions

We will give an operational semantics for the arithmetic expressions given below. We use only three variables X, Y, and Z.

$$\langle e \rangle ::= \langle \text{numseq} \rangle$$
$$| \langle \text{var} \rangle$$
$$| (\langle e \rangle + \langle e \rangle)$$
$$| (\langle e \rangle - \langle e \rangle)$$
$$| (\langle e \rangle * \langle e \rangle)$$
$$\langle \text{var} \rangle ::= X \mid Y \mid Z$$
$$\langle \text{numseq} \rangle ::= \langle \text{num} \rangle \mid \langle \text{numseq} \rangle \langle \text{num} \rangle$$
$$\langle \text{num} \rangle ::= 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$$

(ex.) $(12 + 34)$, $(3 * (45 - X))$ etc.

States

A variable in C is a name for a location. The semantics of a variable in an arithmetic expression is a value stored in the corresponding location.

A state is a function from locations to integers. We use the state with all the values being 0 as the initial state.

For example, under the declarations

`int X = 3; int Y = 4;`

the state is

$\{ (X, 3), (Y, 4), (Z, 0) \}.$

Meta variables

We use meta variables for representing expressions, sequences, integers, variables, and states as in the following.

expressions : $a, a_1, a_2, \dots$

sequences of numbers : $n, n_1, n_2, \dots$

integers : $m, m_1, m_2, \dots$

variables : $x, y, \dots$

states : $\sigma, \sigma_1, \sigma_2, \dots$

Evaluation of arithmetic expressions

We represent the relation that we get an integer m by evaluating an arithmetic expression a in a state σ as follows.

$$\langle a, \sigma \rangle \rightarrow m$$

(ex.)

Suppose $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$. Then the following relations hold.

$$\langle ((10 + 20) * 4), \sigma \rangle \rightarrow 120$$

$$\langle (5 * (X + 1)), \sigma \rangle \rightarrow 20$$

Evaluation of arithmetic expressions

An arithmetic expression $((10 + 20) * 4)$ is evaluated to 120 by evaluating $(10 + 20)$, obtaining 30, and evaluating $(30 * 4)$.

All the arithmetic expressions are evaluated according to some rules.

Evaluation rules for arithmetic expressions

Sequences of numbers

$\langle n, \sigma \rangle \rightarrow m$ (m is an integer represented by the sequence of numbers n .)

Variables

$\langle x, \sigma \rangle \rightarrow \sigma(x)$

Addition

$$\frac{\langle a_1, \sigma \rangle \rightarrow m_1 \quad \langle a_2, \sigma \rangle \rightarrow m_2}{\langle (a_1 + a_2), \sigma \rangle \rightarrow m}$$

(m is the sum of m_1 and m_2 .)

Evaluation rules for arithmetic expressions (Cont.)

Subtraction

$$\frac{\langle a_1, \sigma \rangle \rightarrow m_1 \quad \langle a_2, \sigma \rangle \rightarrow m_2}{\langle (a_1 - a_2), \sigma \rangle \rightarrow m}$$

(m is the difference between m_1 and m_2)

Multiplication

$$\frac{\langle a_1, \sigma \rangle \rightarrow m_1 \quad \langle a_2, \sigma \rangle \rightarrow m_2}{\langle (a_1 * a_2), \sigma \rangle \rightarrow m}$$

(m is the product of m_1 and m_2)

Example 1

Evaluate an arithmetic expression $((10 + 20) * 4)$
under the state $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$.

$$< 10, \sigma > \rightarrow 10 \quad < 20, \sigma > \rightarrow 20$$

$$< (10 + 20), \sigma > \rightarrow 30$$

$$< 4, \sigma > \rightarrow 4$$

$$< ((10 + 20) * 4), \sigma > \rightarrow 120$$

Example 2

Evaluate an arithmetic expression $(5 * (X + 1))$
under the state $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$.

$$\frac{\begin{array}{l} \frac{\langle X, \sigma \rangle \rightarrow 3 \quad \langle 1, \sigma \rangle \rightarrow 1}{\langle (X+1), \sigma \rangle \rightarrow 4} \\ \langle 5, \sigma \rangle \rightarrow 5 \end{array}}{\langle (5 * (X+1)), \sigma \rangle \rightarrow 20}$$

Exercise 1

Evaluate an arithmetic expression

$$((4 + Y) * (5 + Z))$$

in the state $\sigma = \{ (X, 3), (Y, 20), (Z, 13) \}$.

Statements

- We have given semantics to the arithmetic expressions.
- We get values by executing (evaluating) expressions. (In the full set of C, evaluating an expression may change the state.)
- Executing a statement changes the state.

(ex.) $X = 2;$

By executing this statement, the value of X is changed (if the original value is not 2.) Let the state before executing the statement be σ . After executing the statement, the value of X in the state is changed to 2.

Notation concerning states

- We write $\sigma[m / x]$ for the state after assigning an integer m to a variable x in the state σ .

$$(\sigma[m / x])(y) = \begin{cases} m & \text{if } y = x, \\ \sigma(y) & \text{if } y \neq x \end{cases}$$

(ex.) $X = 2;$

By executing the statement in the state σ , the state becomes $\sigma[2 / X]$.

(ex.) $X = (X + 2);$

By executing the statement in the state σ , the state becomes $\sigma[\sigma(X) + 2 / X]$.

Exercise 2

Let $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

Write down all the elements of $\sigma[40 / X]$ in the set notation.

Syntax of statements

We use the statements defined below.

$$\begin{aligned} \langle s \rangle &::= \langle \text{var} \rangle = \langle e \rangle ; \\ &| \quad \langle s \rangle \langle s \rangle \\ &| \quad \text{while } (\langle e \rangle) \{ \langle s \rangle \} \end{aligned}$$

We use meta variables c , c_1 , c_2 , etc. for representing statements.

(Note) Although the C language does not require the body of while statements is surrounded by curly braces, we does require because in our definition a sequence of statements is a statement.

Execution of statements

We write

$$\langle c, \sigma_1 \rangle \rightarrow \sigma_2$$

for representing that executing statement c in state σ_1 terminates in state σ_2 .

(ex.) By executing the statement $Y = 40$; in the state $\{ (X, 10), (Y, 20), (Z, 30) \}$, the state becomes

$$\{ (X, 10), (Y, 40), (Z, 30) \}.$$

We write this relation as follows.

$$\begin{aligned} &\langle Y = 40;, \{ (X, 10), (Y, 20), (Z, 30) \} \rangle \\ &\rightarrow \{ (X, 10), (Y, 40), (Z, 30) \} \end{aligned}$$

Rules for executing statements

Assignments

$$\frac{\langle a, \sigma \rangle \rightarrow m}{\langle x = a;, \sigma \rangle \rightarrow \sigma[m / x]}$$

Sequences of statements

$$\frac{\langle c_1, \sigma \rangle \rightarrow \sigma_1 \quad \langle c_2, \sigma_1 \rangle \rightarrow \sigma_2}{\langle c_1 \ c_2, \sigma \rangle \rightarrow \sigma_2}$$

Example 1

Derive the state after executing the statement
 $Y=40;$

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

$$\frac{\langle 40, \sigma \rangle \rightarrow 40}{\langle Y = 40;;, \sigma \rangle \rightarrow \sigma[40 / Y]}$$

||

$$\{ (X, 10), (Y, 40), (Z, 30) \}$$

Example 2

Derive the state after executing the statement

$X = 3; Y = 40;$

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

$$\frac{\begin{array}{c} < 3, \sigma > \rightarrow 3 \\ \hline < X = 3;, \sigma > \rightarrow \sigma[3/X] \end{array}}{\begin{array}{c} < 40, \sigma[3/X] > \rightarrow 40 \\ \hline < Y = 40;, \sigma[3/X] > \rightarrow (\sigma[3/X])[40/Y] \end{array}} \quad \frac{}{< X = 3; Y = 40; \sigma > \rightarrow (\sigma[3/X])[40/Y]}$$
$$\equiv$$
$$\{ (X, 3), (Y, 40), (Z, 30) \}$$

Exercise 3

Derive the state after executing the statement

$X = (Y + 2); Y = (Y + 3);$

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

Rules for while statements

$$\frac{\langle a, \sigma \rangle \rightarrow 0}{\langle \text{while } (a) \{ c \}, \sigma \rangle \rightarrow \sigma}$$

$$\frac{\langle a, \sigma \rangle \rightarrow m \quad \langle c, \sigma \rangle \rightarrow \sigma_1 \quad \langle \text{while } (a) \{ c \}, \sigma_1 \rangle \rightarrow \sigma_2}{\langle \text{while } (a) \{ c \}, \sigma \rangle \rightarrow \sigma_2} \text{ if } m \neq 0$$

Example 3

Derive the state after executing the statement

while (Y) { Y = (Y - 20); }

in the state $\sigma = \{ (X, 10), (Y, 20), (Z, 30) \}$.

$$\begin{array}{c}
 \frac{\langle Y, \sigma \rangle \rightarrow 20 \quad \langle 20, \sigma \rangle \rightarrow 20}{\langle (Y-20), \sigma \rangle \rightarrow 0} \quad \frac{\langle Y, \sigma[0/Y] \rangle \rightarrow 0}{\langle \text{while}(Y)\{Y=(Y-20);\}, \sigma[0/Y] \rangle \rightarrow \sigma[0/Y]} \\
 \frac{\langle Y, \sigma \rangle \rightarrow 20 \quad \langle Y=(Y-20);, \sigma \rangle \rightarrow \sigma[0/Y]}{\langle \text{while}(Y)\{Y=(Y-20);\}, \sigma \rangle \rightarrow \sigma[0/Y]} \\
 \parallel \\
 \{ (X, 10), (Y, 0), (Z, 30) \}
 \end{array}$$

Exercise 4

Derive the state after executing the statement

`while ( Y ) { Y = (Y - 20); }`

in the state $\sigma = \{ (X, 10), (Y, 40), (Z, 30) \}$.